AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 1 · STUDENT LESSON

Representing inputs as vectors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Before you start

A parcel classifier uses width and height to predict a class. Why must its final class label stay outside the input features?

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Encode a record using a fixed feature order and units. Try the recall before opening notes.

Your learning target

I can encode a record using a fixed feature order and units.

I can distinguish a **vector dimension** from the number of examples.

I can detect missing values and target leakage before computation.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from ML Builder Expo. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Encode a record using a fixed feature order and units

Distinguish a **vector dimension** from the number of examples

Detect missing values and target leakage before computation

Retrieve what you already know

> A parcel classifier uses width and height to predict a class. Why must its final class label stay outside the **input** features?

My starting explanation

This week's end goal

By the end, I can explain representing **input**s as **vector**s using a traced example and a stated limitation.

Notice the evidence

> Investigate the supplied Representing inputs as vectors case. Separate observations, given facts and assumptions before choosing an explanation.

A neural network receives numbers in a specified order. Before learning how it computes, we need to represent one example without losing the meaning of its fields.

A **vector** is an ordered list: [4, 2] can mean width 4 cm and height 2 cm if that is the declared order. Reversing the two numbers describes a different rectangle under the same **schema**.

For this fictional parcel investigation, use [width in cm, height in cm]. Width and height must both be present and greater than zero. The **vector** has **dimension** 2 because it contains two **component**s.

Record IDs identify examples; they do not measure the parcel. The target class is stored separately.

Record	Width and height in cm	Input vector
A	4 and 2	[4, 2]
B	2 and 4	[2, 4]
C	3 and 3	[3, 3]

There are three examples, each with two features. Stacking their **vector**s makes a table with three rows and two columns. Do not confuse the number of rows with the **dimension** of one **input**.

> **List the given facts and one fact that is not supplied.**

See how it works

> Follow the mechanism for Representing inputs as vectors. Connect each labelled input, deciding step and result.

Read fields by name, convert to the declared units, validate, and only then write the **vector**. Parcel D has width 50 mm and height 30 mm. Because 10 mm equals 1 cm, its encoded **vector** is [5, 3].

Writing [50, 30] would preserve the numbers while changing the physical meaning by a factor of ten.

```
def encode(width_mm, height_mm):
    if width_mm is None or height_mm is None:
        raise ValueError("Missing measurement")
    if width_mm <= 0 or height_mm <= 0:
        raise ValueError("Measurements must be positive")
    return [width_mm / 10, height_mm / 10]

print(encode(50, 30))  # [5.0, 3.0]
```

This short function assumes each supplied value is a number in mm or None for missing. It is a classroom example, not a complete **input** validator. Trace the conversion before running it.

The function represents data; it does not fit a model or predict a class.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

Compare the cases

> Compare the supplied cases for Representing inputs as vectors. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Swap only A's **vector** order before **prediction**.

Predicted result

Prediction: The encoded **input** becomes [7, 2], numerically identical to B's original **input**.

Why the result follows

Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Representing inputs as vectors to explain a concrete decision, rather

> than only memorising definitions.

Word	Meaning
Vector	An ordered list of numerical components
Component	One value at a specified position in a vector
Dimension	The number of components in a vector
Schema	The recorded feature order, units and validity rules

> **Use two terms to explain a step in this week's example.**

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Representing inputs as vectors, predict a step, then compare with its explanation.

An input vector keeps an agreed order

SOURCE RECORD	ENCODING ACTION	MODEL INPUT
D width 50 mm	Divide width by 10	First component 5
D height 30 mm	Divide height by 10	Second component 3
Schema width then height	Keep named positions	Vector [5, 3] in cm

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Parcel E has width 20 mm and height 70 mm. Encode it and explain why [7, 2] is not the same input under this schema.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Representing inputs as vectors

Encode the same pixel measurements in a fixed **vector** order.

The facts we will use

Supplied facts

Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/1

Representing inputs as vectors

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare coordinate order

Coordinate 0 is left brightness; coordinate 1 is right brightness.

2. Encode both images

A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.

3. Check input contract

Use the same order when training and predicting; record missing measurements rather than guessing.

CHANGE ONE THING

Swap only A's vector order before prediction.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare coordinate order	Coordinate 0 is left brightness; coordinate 1 is right brightness.
2. Encode both images	A becomes [2, 7]; B becomes [7, 2]. Their sums match but their spatial patterns differ.
3. Check input contract	Use the same order when training and predicting; record missing measurements rather than guessing.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

What this example does not establish

Keep the stated limit

These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

Keep the stated limit

Read the labelled picture: Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Swap only A's **vector** order before **prediction**. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Representing inputs as vectors claim and explain a repair using the supplied evidence.

Check the claim against the facts

A missing height is unknown, not zero. Under this **schema** zero is also invalid, so reject the incomplete record and request the measurement.

A different project might use a documented **training**-fitted replacement plus a missingness flag; that would be a different **schema** that must be applied consistently.

Check the claim against the facts

A changed field order must also change every consumer that reads positions. A **vector** can have the expected length and still be wrong because its order, units or meaning changed.

A class label appended as a third **component** causes target leakage even if the software accepts three values.

Locate the earliest unsupported step

Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this **schema**.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

Practise Representing inputs as vectors with support, then explain the deciding step in your own words.

Write A, B and C as ordered pairs from the lesson. State the **dimension** of each and the shape of the stacked table.

Parcel E has width 20 mm and height 70 mm. Encode it and explain why [7, 2] is not the same input under this schema.

Explain your prediction to a partner before comparing results.

Practise with less help: Convert before encoding

Use workbook W2 for the connected example “Representing **input**s as **vector**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Representing **input**s as **vector**s. Proposed explanation: “Order does not matter”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Representing inputs as vectors. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A new record has width 60 mm and height 10 mm. Give its **vector** in the agreed cm **schema**, and name one error in [60, 10].

```
def encode(width_mm, height_mm):
    if width_mm is None or height_mm is None:
        raise ValueError("Missing measurement")
    if width_mm <= 0 or height_mm <= 0:
        raise ValueError("Measurements must be positive")
    return [width_mm / 10, height_mm / 10]

print(encode(50, 30))  # [5.0, 3.0]
```

Record	Your evidence
Given input and fixed rule	
Prediction before checking	

Record	Your evidence
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Representing inputs as vectors. Explain why each step is justified.

Explain → apply → challenge

Write A, B and C as ordered pairs from the lesson. State the **dimension** of each and the shape of the stacked table.

Width is 4 cm and height is missing. A teammate writes [4, 0]. Explain why this is invalid and give the action required by this **schema**.

Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two **input**s apart.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled representing **input**s as **vector**s **evidence** sheet to Image-pattern Network Investigation. Show how you can encode a record using a fixed feature order and units.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Two parcels can share width and height yet differ in material or weight. Their **vector**s are identical in this representation, so a deterministic model receiving only these features cannot distinguish them.

Adding a feature changes the **schema** and **dimension**; it does not automatically improve **prediction**s. Keep data splits and evaluation discipline from Level 4 (Grade 9).

> **Two records both encode as [4, 2] but have different target classes. What can this representation reveal, and what can it not reveal?**

Next week, we will multiply each **component** by a matching weight and add the results. The computation is meaningful only if the feature order stays fixed.

Two parcels have the same width and height but different material and target class. Explain why adding more examples alone cannot let this deterministic two-feature model tell these two **input**s apart.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Representing inputs as vectors case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

An input vector keeps an agreed order

SOURCE RECORD	ENCODING ACTION	MODEL INPUT
D width 50 mm	Divide width by 10	First component 5
D height 30 mm	Divide height by 10	Second component 3
Schema width then height	Keep named positions	Vector [5, 3] in cm

Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new schema independently

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

A new toy battery task uses [temperature in degrees C, voltage in V, age in days]. Record K has voltage 5, age 12, temperature 18. Record L has temperature 20, voltage 4, age 0.

Encode both; state the **vector dimension** and table shape. Age zero is valid. The future failure label is also supplied in the data file: should it be appended?

> **My independent procedure, result and limitation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

Your lesson route

> Use this map to navigate Representing inputs as vectors, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Encode the same pixel measurements in a fixed **vector** order.

By the end, I can	How I will show it
Encode a record using a fixed feature order and units	Explain the deciding step, show a trace or test, and state one limit.
Distinguish a vector dimension from the number of examples	Explain the deciding step, show a trace or test, and state one limit.
Detect missing values and target leakage before computation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Representing inputs as vectors without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Encode a record using a fixed feature order and units

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Encode the same pixel measurements in a fixed **vector** order.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Use the same order when **training** and predicting; record missing measurements rather than guessing.

1. Declare coordinate order

2. Check **input** contract

3. Encode both images

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check **input** contract Use the same order when **training** and predicting; record missing measurements rather than guessing.

A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Swap only A's **vector** order before **prediction**. Which conclusion is supported by the supplied facts?

1. The encoded **input** becomes [7, 2], numerically identical to B's original **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Changing coordinate order changes feature meaning even if the numbers are unchanged.

Repair a missing building block

Review ML Builder Expo (ai-4/36). Goal: Defend the project using traceable **evidence**. Short practice: Which three supporting items should be visible at the expo?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

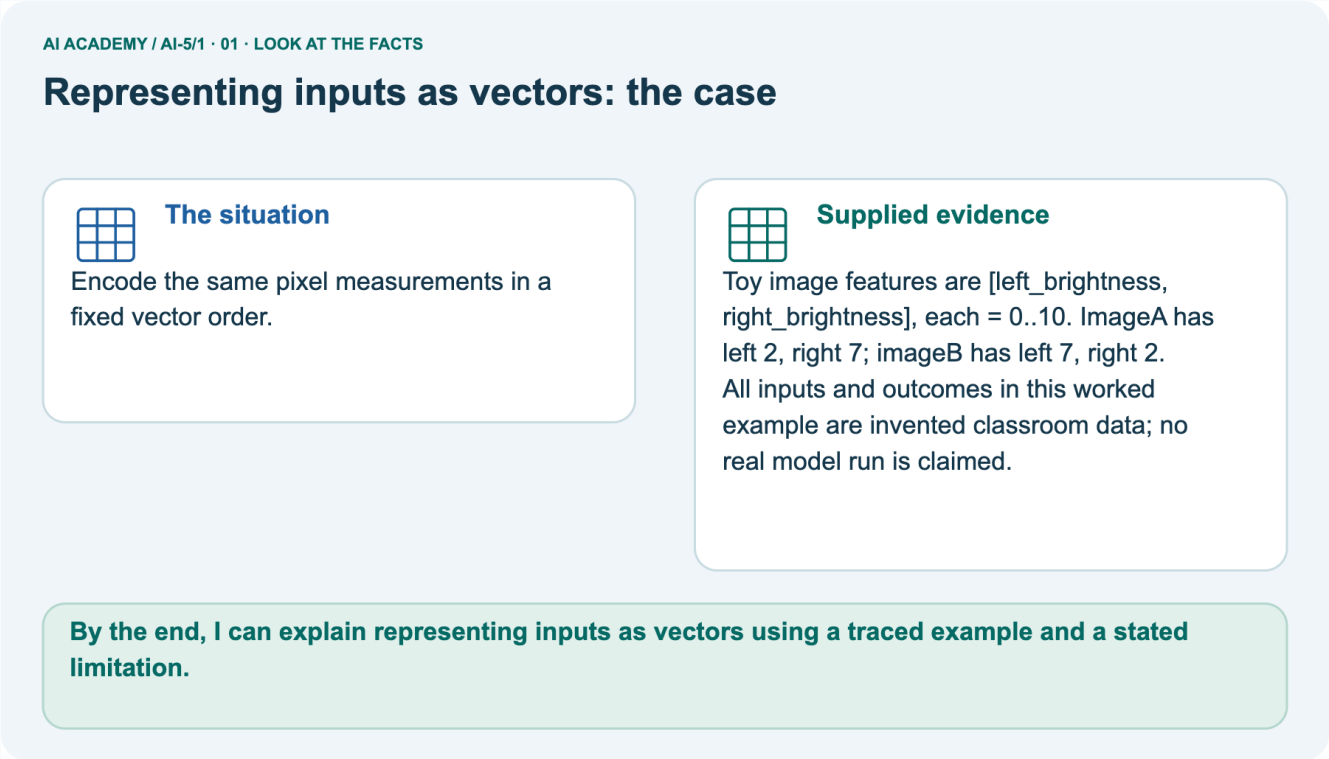
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Representing inputs as vectors graphic using its labels, deciding evidence and relevant limitation.

01. Representing inputs as vectors: the case



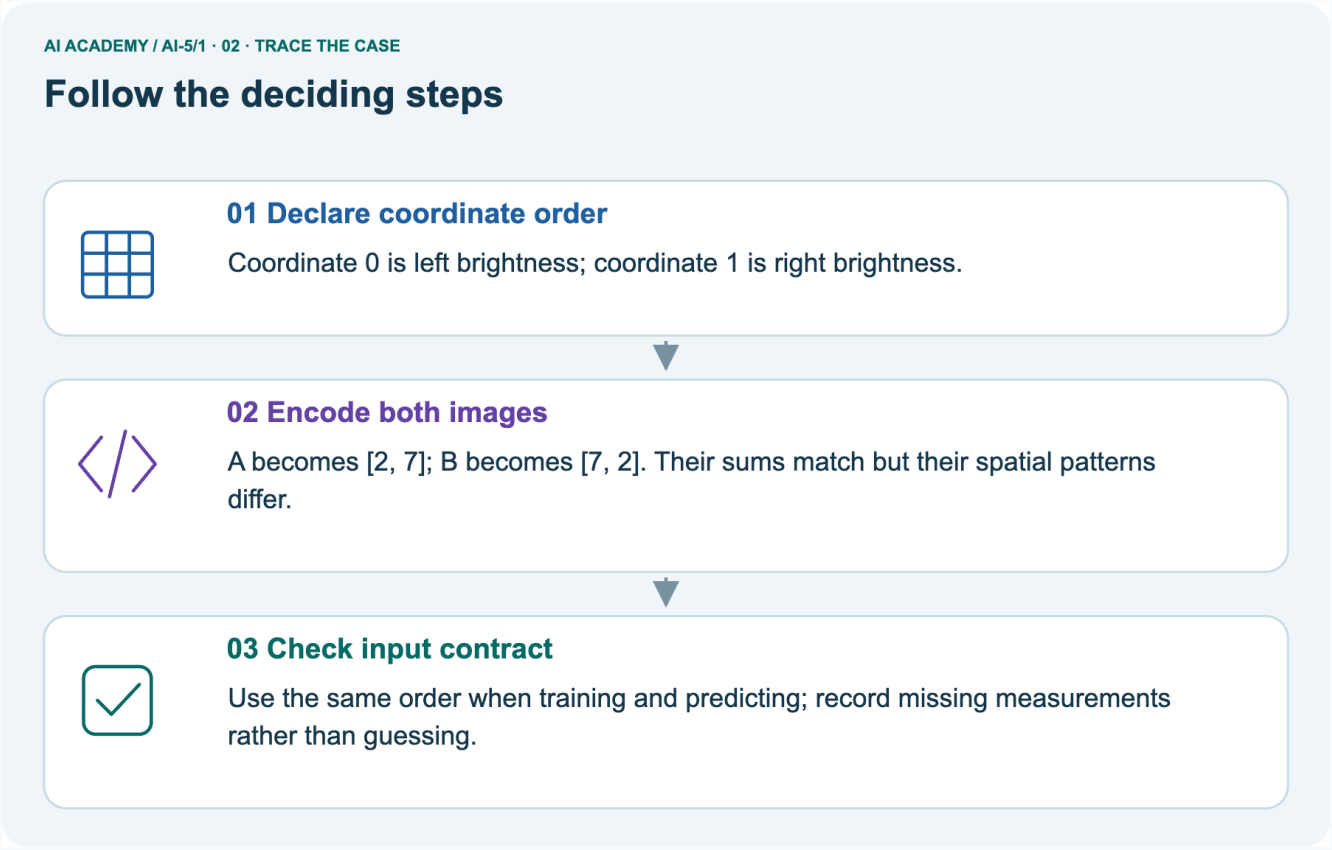
Follow the labelled picture with the matching explanation. Open the vector diagram

Encode the same pixel measurements in a fixed **vector** order. Supplied facts: Toy image features are [left_brightness, right_brightness], each = 0..10. ImageA has left 2, right 7; imageB has left 7, right 2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Declare coordinate order: Coordinate 0 is left brightness; coordinate 1 is right brightness. → Encode both images: A becomes [2, 7]; B becomes [7, 2].

Their sums match but their spatial patterns differ. → Check **input** contract: Use the same order when **training** and predicting; record missing measurements rather than guessing.

> Point to each numbered stage and explain which recorded **input** it uses.

03. An input vector keeps an agreed order

AI ACADEMY / AI-5/1 · 03 · INSPECT THE LESSON MODEL

An input vector keeps an agreed order

Source record	Encoding action	Model input
D width 50 mm	Divide width by 10	First component 5
D height 30 mm	Divide height by 10	Second component 3
Schema width then height	Keep named positions	Vector [5, 3] in cm

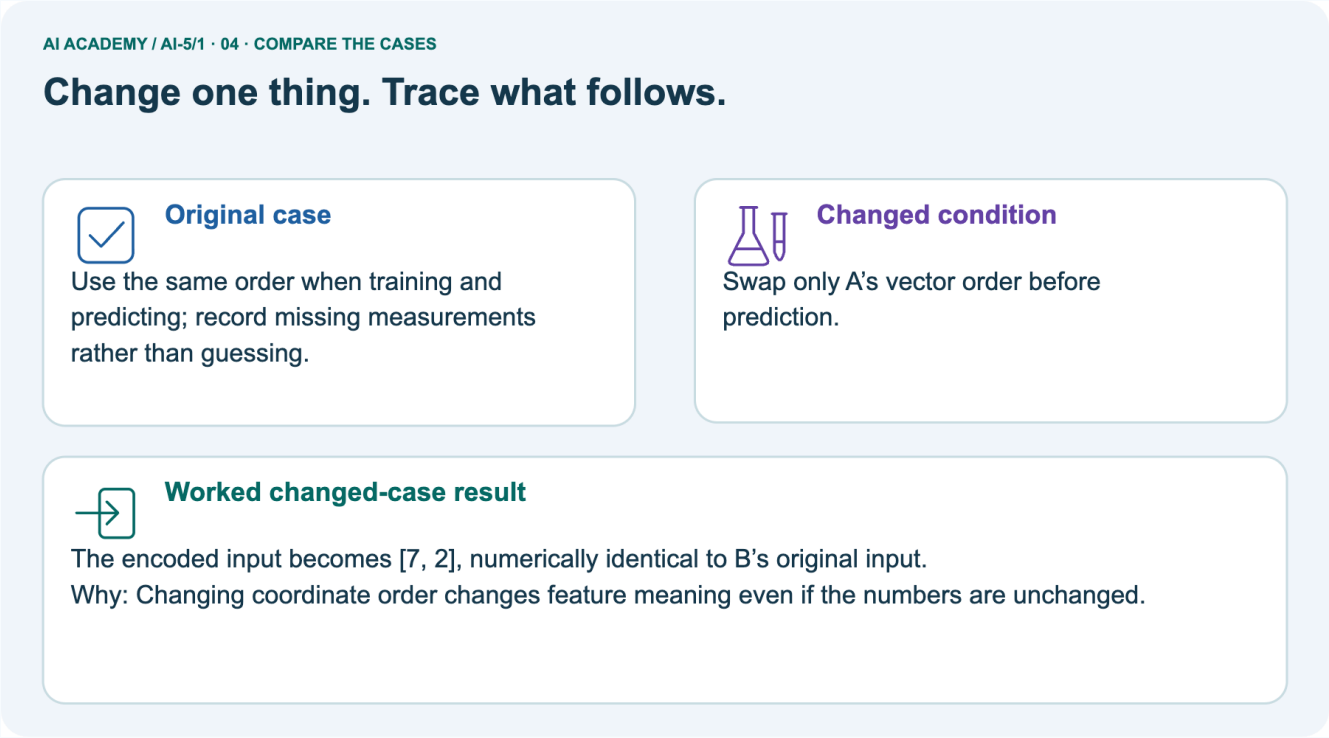
Read each row left to right, then assemble the final ordered pair. The schema explains the numbers; the vector alone does not.

Follow the labelled picture with the matching explanation. Open the vector diagram

An **input vector** keeps an agreed order. D width 50 mm | Divide width by 10 | First **component** 5. D height 30 mm | Divide height by 10 | Second **component** 3. **Schema** width then height | Keep named positions | **Vector** [5, 3] in cm

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



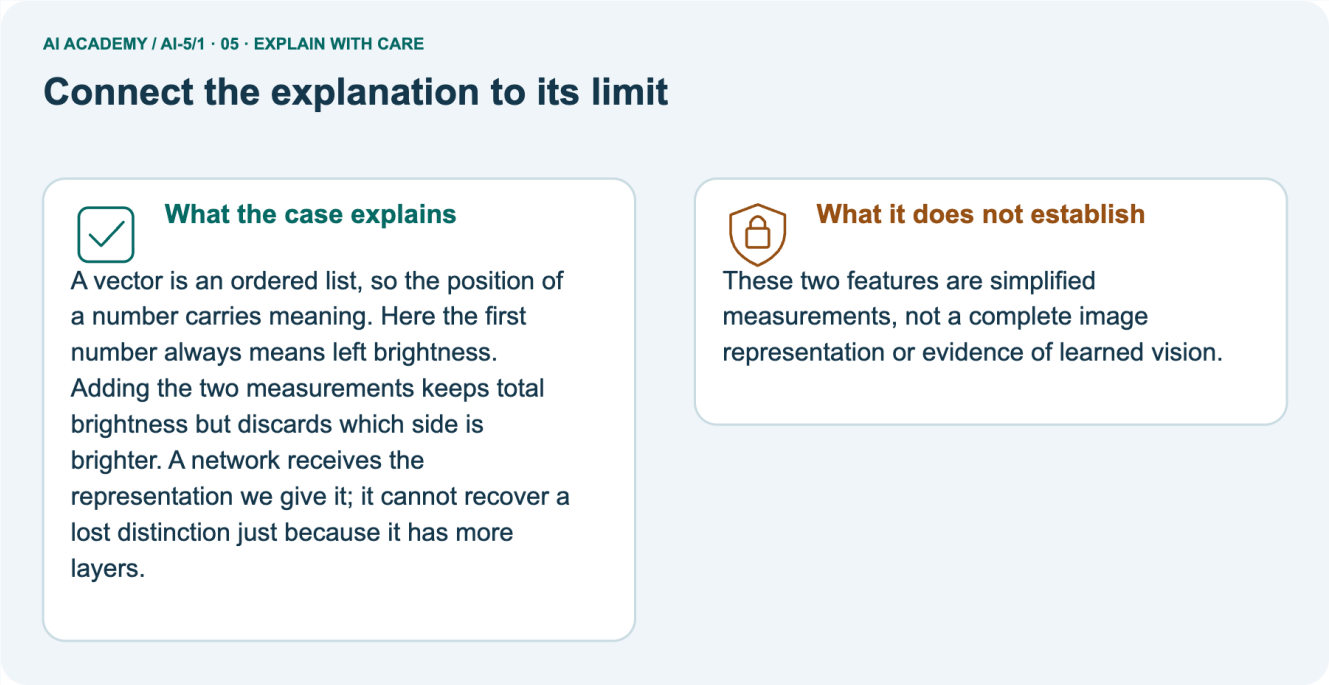
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use the same order when **training** and predicting; record missing measurements rather than guessing. Changed condition: Swap only A's **vector** order before **prediction**.

Expected result: The encoded **input** becomes [7, 2], numerically identical to B's original **input**. Reason: Changing coordinate order changes feature meaning even if the numbers are unchanged.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

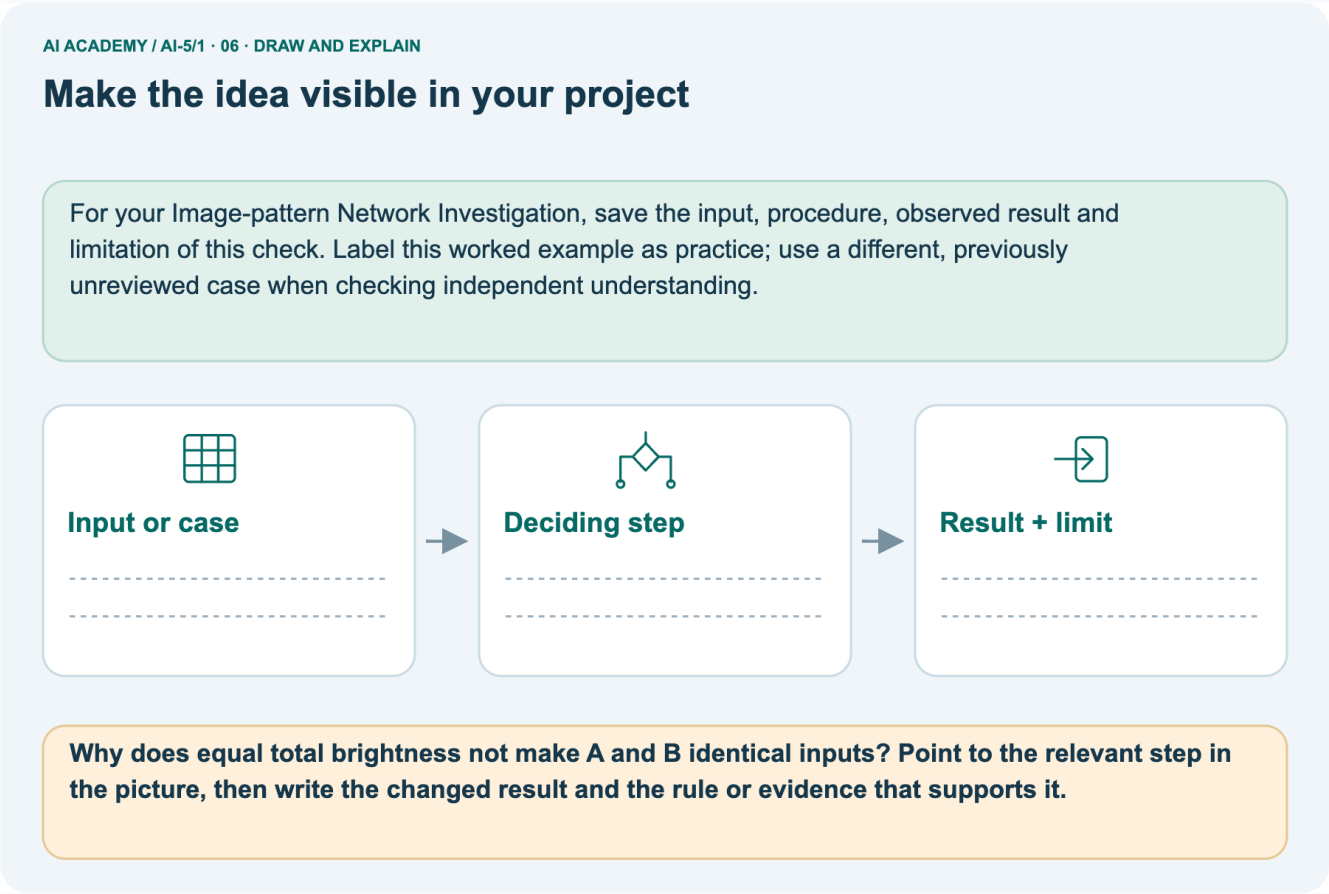
A **vector** is an ordered list, so the position of a number carries meaning. Here the first number always means left brightness. Adding the two measurements keeps total brightness but discards which side is brighter.

A network receives the representation we give it; it cannot recover a lost distinction just because it has more layers.

Limit: These two features are simplified measurements, not a complete image representation or **evidence** of learned vision.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why does equal total brightness not make A and B identical **input**s? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.